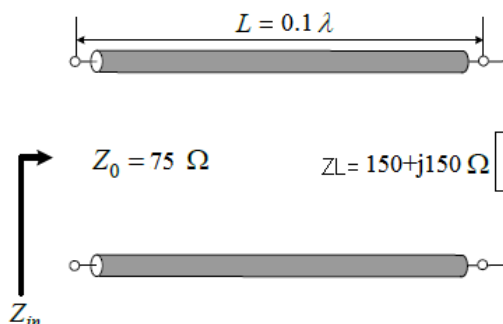


CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. Examination (EC)****December 2016****EC402.01 RF & Microwave Engineering****Date: 09.12.2016, Wednesday****Time: 1.30 p.m. to 4.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Attach given smith chart with main answer book in appropriate section.

Q-1 Answer the followings

- (a) A loss less line of characteristic impedance Z_0 is terminated in pure reactance of $-jZ_0$ value. VSWR is _____. **1**
- (b) A 10 km long line has a characteristic impedance of 400Ω . If line length is 100 km, the characteristic impedance is _____. **1**
- (c) A line of characteristic impedance 400Ω is terminated by a resistance of 400Ω . The reflected quantities are _____. **1**
- (d) Voltage reflection coefficient is the ratio of _____ wave to _____ wave. **1**
- (e) If the short circuited and open circuit impedance of line are 5 and 20Ω respectively the find characteristics impedance. **1**
- (f) A loss less line has a characteristic impedance of 50Ω and C equal to 40 pF/m . find the inductance of line. **1**
- (g) An isolator is a two _____ device which produces a minimum attenuation to wave propagation in one direction and very high attenuation in the opposite direction. (reciprocal/ non reciprocal) **1**
- Q-2 (a)** Why TEM mode cannot propagate in Waveguide? **2**
- (b)** Use Smith chart to find the following quantities for the transmission-line circuit shown in the Fig 1. **10**
- (a) SWR and draw SWR circle on the line.
 - (b) The reflection coefficient at the load.
 - (c) The input impedance of the line.
 - (d) The distance from the load to the first voltage minimum.
 - (e) The distance from the load to the first voltage maximum.

**Figure 1****OR**

- (b) Derive the wave equation with complete solutions regarding wave propagation in **10**

	rectangular waveguide for TM mode.	
(c)	An air filled hollow rectangular waveguide has cross sections of $22.86 \times 10.16 \text{ cm}^2$. Calculate cutoff frequency for dominant mode.	2
	OR	
(c)	Design a quarter wave transmission transformer to match a line having impedance of 35Ω , feeding at 3MHz to a load of 50Ω	2
Q-3 (a)	Explain standing waves	2
(b)(1)	Based on distribution theory of transmission line, draw equivalent circuit of transmission line and equation of voltage and current for lossless transmission line.	6
(2)	A rectangular air filled waveguide with dimension $2.28 \times 1.01 \text{ cm}^2$ is operated with a dominant mode $\text{TE}_{1,0}$ with frequency 9.2 GHz. Find f_c , V_p , λ_g	6
	OR	
(b)(1)	Explain operation of circulator with its S matrix.	6
(2)	Explain operation of magic tee with its S matrix.	6
	Section –II	
Q-4	Answer the followings	
(a)	Waveguide is similar to _____ filter?	1
(b)	A dominant mode of the rectangular waveguide in TE mode is _____	1
(c)	In a loss less line the propagation constant is equal to _____	1
(d)	Modes have the same cut off frequencies but different field distributions are called _____ modes in waveguide.	1
(e)	TE_{10} mode has the _____ cut off frequency for that it is dominant mode in a rectangular waveguide. (lowest/ highest/ moderate).	1
(f)	A resistive microwave load with $Z_L = 150 \Omega$ is connected to 50Ω coaxial line. SWR is _____.	1
(g)	Which structure in the TWT delays the forward progress of the traveling wave?	1
Q- 5 (a)	Explain in detail electron bunching process with help of diagram.	2
(b) (1)	Explain energy band gap diagram and I-V characteristics of Schottky barrier diode.	6
(2)	What is Gunn effect? Explain I-V characteristics of Gunn diode.	6
	OR	
(b) (1)	Explain reflex klystron oscillator and its mode characteristics in detail.	6
(2)	A microstrip line has following parameters $\epsilon_r=5.23$, $h=0.01778 \text{ mm}$, $t=0.007112 \text{ mm}$, $w=0.254 \text{ mm}$, calculate z_0 .	2
(3)	A transmission line has characteristic impedance of 50Ω and is terminated in a load impedance of 75Ω . Calculate reflection coefficient and standing wave ration.	4
Q- 6(a)	What is frequency jumping in magnetron?	2
(b) (1)	Explain microstrip line structure and losses associated with microstrip line.	6
(2)	Draw appropriate diagram of TWTA and explain amplification process along with diagram.	6
	OR	
(b) (1)	Explain faraday rotation principle and working of isolator.	6
(2)	Explain directional coupler and its s matrix in detail.	6
